

1. What is Middleware & `next()` ?

Middleware euta yesto function ho jasle **Request (req)** ra **Response (res)** ko bichma basera kaam garchha.

[Client Request] —> [Middleware 1] —> [Middleware 2] —> [Route Controller / Final Response]

Harek middleware sanga 3 veta parameters hunchhan: `req`, `res`, ra `next`.

- **`next()` function:** Yo ekdamai important chha. Jaba middleware le aafno kaam sakchha, yesle `next()` lai call garnaiparchha. `next()` call nagaresi request agadi baddhai na (browser loading gari matra rahanchha).

2. Built-in Middleware

Express framework sangai aafai aune software layers haru yeta chhan:

`express.json()`

- **Kaam:** Client le POST/PUT request ma pathako JSON data (body) lai read (parse) garera `req.body` bhitra halne kaam garchha.
- **Usage:** `app.use(express.json());`

`express.urlencoded()`

- **Kaam:** HTML Form (jastai login/register form submits) bata aune data parse garna use hunchha.
- **Usage:** `app.use(express.urlencoded({ extended: true }));`

Static Files (`express.static`)

- **Kaam:** Images, CSS, ba Front-end ko HTML files haru lai direct browser ma open garna access dinchha (jastai image files display garna).
- **Usage:** `app.use(express.static('public'));` (Yesle `public` folder ko items public banauchha).

3. Global Middleware vs Route-Specific Middleware

- **Global Middleware:** App ko harek endpoint (URL) run huda yo automatic chalthha. `app.use(middlewareName)` garera halinchha.
 - **Route-Specific Middleware:** Kunai euta specific route (jastai dashboard checkout) ma matra use hunchha.
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4. Custom Middlewares (Practical Examples)

Aba hami aafai custom middleware banayera herū:

A. Logger Middleware (Global Example)

Harek patak user le kun endpoint hit garyo ra kun time ma garyo bhanera terminal ma record rakhna:

javascript

```
const loggerMiddleware = (req, res, next) => {
  const currentMethod = req.method; // GET, POST etc.
  const currentUrl = req.url;        // /api/products
  const timestamp = new Date().toISOString();

  console.log(`[${timestamp}] ${currentMethod} request received for ${currentUrl}`);

  next(); // Request lai agadi badhauna mandatory chha!
};

// Global banauko lagi main file ko mathi nai yesari thapne:
app.use(loggerMiddleware);
Use code with caution.
```

B. Authentication Middleware (Route-Specific)

Tyo user login chha ki chhaina check garna. Login nabhai dashboard pathauna mildaina.

javascript

```
const checkAuth = (req, res, next) => {
  // Manau client le headers ma secret token pathaudai chha
  const token = req.headers['authorization'];
```

```

    if (token === 'valid-token-123') {
      console.log("User successfully logged in!");
      next(); // Token match bhaye dashboard jana dine
    } else {
      // Token milena bhane response yehi bata block garera firta
      pathaune
      res.status(401).json({ error: "Access Denied! Please login first."
    });
  }
};
Use code with caution.

```

C. Authorization Middleware (Role Check)

User login ta chha, tara ke tyo **Admin** ho? Normal user le product delete garna paudaena.

javascript

```

const checkAdmin = (req, res, next) => {
  // Manau auth middleware le user detail 'req.user' ma pass gardyo
  const userRole = req.headers['user-role']; // Client le pathako
  standard role

  if (userRole === 'admin') {
    next(); // Admin ho bhane delete garna dine
  } else {
    res.status(403).json({ error: "Forbidden! Admin power matra
    chahinchha yesko lagi." });
  }
};
Use code with caution.

```

5. Complete Implementation code (Putting It Together)

Aba yo sabai middleware haru 1 thau ma kasari use garchhan, full architecture herũ:

javascript

```

const express = require('express');
const app = express();
const PORT = 3000;

// 1. Built-in Middlewares (Global)

```

```

app.use(express.json());
app.use(express.static('public'));

// 2. Custom Global Middleware (Logger)
app.use((req, res, next) => {
  console.log(`Req Path: ${req.path} | Time: ${Date.now()}`);
  next();
});

// Dummy Route - Publicly sabaile herna paune
app.get('/api/free-content', (req, res) => {
  res.send("Yo sabaile herna milne public data ho.");
});

// 3. Protected Route (Auth ra Role check haleko)
// checkAuth ra checkAdmin cross bhae matra final callback run hunchha
const checkAuth = (req, res, next) => {
  const authHeader = req.headers['authorization'];
  if(authHeader === 'secret') next();
  else res.status(401).json({ message: "Unauthorized" });
};

const checkAdmin = (req, res, next) => {
  const role = req.headers['role'];
  if(role === 'admin') next();
  else res.status(403).json({ message: "Admin only area!" });
};

// Route-specific implementations
app.delete('/api/delete-database', checkAuth, checkAdmin, (req, res) => {
  res.json({ message: "Database deleted successfully by Admin!" });
});

app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});

```

Use code with caution.

Key Rules to Remember:

1. Middleware run huda **Order (क्रम)** ekdam matter garchha. `express.json()` route bhandha mathi hunaiparchha, natra parsing hudaina.
2. Harek custom middleware ko end ma `next()` thichna nabhulnu hola, natra backend freeze hunchha.
3. Middleware batai error response handle garera process chunk end garna pani sakinchna (`res.status().json()`).

